

## **Subject Description Form**

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| <b>Subject Code</b>                                   | EE537                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Subject Title</b>                                  | Railway Vehicles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Level</b>                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Collaboration<br/>Institute</b>                    | MTR Academy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To enable students to describe the design and applications of railway vehicles.</li> <li>2. To enable the students to describe the current state-of-the-art on the design, operation and maintenance of railway vehicles in Hong Kong and overseas.</li> <li>3. To enable students to present the procurement process of railway vehicles and the necessary management.</li> <li>4. To enable the students to conduct modelling and analysis on the components in railway vehicles.</li> <li>5. To enable the students to describe the testing standards for vehicles; and the inspection and quality control measures.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Intended Learning<br/>Outcomes</b>                 | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a. Identify various types and configurations of railway vehicles.</li> <li>b. Discuss the design principles and system performance of railway vehicles and be aware of the latest developments in the technology.</li> <li>c. Elaborate on the project management process for railway vehicle procurement and devise feasibility study and maintenance planning.</li> <li>d. Apply appropriate modelling for vehicles, body design and train dynamics in vehicle performance analysis.</li> <li>e. Given the acceptance standards, formulate tests and inspections for quality control purposes.</li> <li>f. Appreciate the role of engineers on matters other than technical issues.</li> <li>g. Recognise the importance to engage in self-learning on the latest technologies on railway vehicle design at this advanced level of study.</li> </ol>                                                                                                                                                            |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. <b><i>Project management for procurement of railway vehicle:</i></b> Planning and preliminary design, System selection, definition of vehicle, specification, design management, testing and commissioning, maintenance planning.</li> <li>2. <b><i>Railway vehicle design and development:</i></b> Types and configurations of railway vehicles, design principles, system performance, Interface and environmental considerations, modern development.</li> <li>3. <b><i>System description and mechanism design:</i></b> Carbody, bogie, coupler, door, brake, pneumatics, air-conditioning, traction and control, pantograph, and train management system.</li> <li>4. <b><i>Vehicle modelling and gauging:</i></b> Rail vehicle components, suspension system, modelling of vehicles and analysis, kinetic envelope, load gauge.</li> <li>5. <b><i>Vehicle structures and dynamics:</i></b> Body shell design, load cases, structural testing and analysis, fundamentals of train dynamics, wheel rail interface, track geometry effect, derailment prediction.</li> </ol> |

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|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|---|---|----------|---|---|---|
|                                                                        | 6. <b>Vehicle acceptance and testing:</b> Acceptance standards, type test, inspection and quality control, static testing, dynamic runs, trial operation and reliability monitoring.<br><br><b>Case Study:</b><br>Site Visits to MTRCL Depots<br>Industrial/Research Seminars                                                            |             |                                                   |   |   |          |   |   |   |
| <b>Teaching/Learning Methodology</b>                                   | The main lecturers are from MTRC, and their experiences/knowledge are shared with students via lectures and tutorials for conveying the concept and theories. The site visit to MTR system has reinforced the pragmatic design and application in a realistic system. Problem solving skill and team work are trained via minor project. |             |                                                   |   |   |          |   |   |   |
|                                                                        | Teaching/Learning Methodology                                                                                                                                                                                                                                                                                                            | Outcomes    |                                                   |   |   |          |   |   |   |
|                                                                        |                                                                                                                                                                                                                                                                                                                                          | a           | b                                                 | c | d | e        | f | g |   |
|                                                                        |                                                                                                                                                                                                                                                                                                                                          | Lectures    | ✓                                                 | ✓ | ✓ | ✓        | ✓ | ✓ | ✓ |
| Tutorials                                                              |                                                                                                                                                                                                                                                                                                                                          | ✓           | ✓                                                 | ✓ | ✓ | ✓        | ✓ |   |   |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b> | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                        | % weighting | Intended subject learning outcomes to be assessed |   |   |          |   |   |   |
|                                                                        |                                                                                                                                                                                                                                                                                                                                          |             | a                                                 | b | c | d        | e | f | g |
|                                                                        | 1. Examination                                                                                                                                                                                                                                                                                                                           | 60%         | ✓                                                 | ✓ | ✓ | ✓        | ✓ |   |   |
|                                                                        | 2. Test                                                                                                                                                                                                                                                                                                                                  | 25%         | ✓                                                 |   | ✓ | ✓        | ✓ | ✓ |   |
|                                                                        | 3. Presentation with Essay Submission                                                                                                                                                                                                                                                                                                    | 15%         | ✓                                                 | ✓ | ✓ | ✓        | ✓ | ✓ | ✓ |
|                                                                        | Total                                                                                                                                                                                                                                                                                                                                    | 100%        |                                                   |   |   |          |   |   |   |
|                                                                        | The outcomes on concepts, design and applications are assessed by the usual means of examination and test.<br><br>The problem solving skill is evaluated via presentation (with essay submission).                                                                                                                                       |             |                                                   |   |   |          |   |   |   |
| <b>Student Study Effort Expected</b>                                   | Class contact:                                                                                                                                                                                                                                                                                                                           |             |                                                   |   |   |          |   |   |   |
|                                                                        | ▪ Lecture/Tutorial                                                                                                                                                                                                                                                                                                                       |             |                                                   |   |   | 33 Hrs.  |   |   |   |
|                                                                        | ▪ Presentation seminar                                                                                                                                                                                                                                                                                                                   |             |                                                   |   |   | 3 Hrs.   |   |   |   |
|                                                                        | ▪ Site visit                                                                                                                                                                                                                                                                                                                             |             |                                                   |   |   | 3 Hrs.   |   |   |   |
|                                                                        | Other student study effort:                                                                                                                                                                                                                                                                                                              |             |                                                   |   |   |          |   |   |   |
|                                                                        | ▪ Presentation preparation/report                                                                                                                                                                                                                                                                                                        |             |                                                   |   |   | 24 Hrs.  |   |   |   |
|                                                                        | ▪ Self-study                                                                                                                                                                                                                                                                                                                             |             |                                                   |   |   | 42 Hrs.  |   |   |   |
|                                                                        | Total student study effort                                                                                                                                                                                                                                                                                                               |             |                                                   |   |   | 105 Hrs. |   |   |   |
| <b>Reading List and References</b>                                     | <b>Textbooks:</b><br>1. A.H. Wickens, Fundamentals of Rail Vehicle Dynamics: Guidance and Stability, Swets & Zeitlinger Publishers, 2003                                                                                                                                                                                                 |             |                                                   |   |   |          |   |   |   |
|                                                                        | <b>Reference books:</b><br>1. Selected papers from the Proceedings of IMechE Part F – Journal of Rail and Rapid Transit                                                                                                                                                                                                                  |             |                                                   |   |   |          |   |   |   |